Ridge Regression

- By retaining a subset of the predictors and discarding the rest, subset selection produces a model that is interpretable and with possibly lower prediction error than the full model.
- However, because it is a discrete process, it often exhibits high variance. Shrinkage methods such as Ridge regression are more continuous and don't suffer as much from high variability.
- Ridge regression shrinks the regression coefficients by imposing a penalty on their size:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \arg\min_{\beta} \sum_{i=1}^{N} \left(y_i - \beta_0 - \sum_{h=1}^{p} x_{ih}\beta_h \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2, \tag{$*$}$$

where $\lambda \geq 0$ is a complexity parameter controlling the amount of shrinkage; the larger λ is, the greater the amount of shrinkage.

- Note that the expression in $(*)$ is the Lagrangean associated with the following optimization problem:

$$\min_{\beta} \sum_{i=1}^{N} \left(y_i - x_i'\beta \right)^2 \quad \text{s.t.} \quad \sum_{h=1}^{p} \beta_h \leq t, \tag{$**$}$$

making explicit the size constraint on the parameters.

- When there are many correlated variables in a linear regression model, their coefficients can become poorly determined and exhibit high variance. Then, a wildly large positive coefficient on one variable can be cancelled by a large negative coefficient on a correlated cousin. The size constraint in $(**)$ alleviates this problem.

- We can rewrite the criterion in $(*)$ in matrix notation:

$$\mathrm{RSS}(\lambda) = (\underline{y} - X\beta)'(\underline{y} - X\beta) + \lambda\beta'\beta, \quad \text{minimizing:} \tag{1}$$

$$-X'\underline{y} - X'\underline{y} + X'X\hat{\beta} + X'X\hat{\beta} + \lambda 2\hat{\beta} = 0. \tag{2}$$

$$\Rightarrow (X'X + \lambda I_p)\hat{\beta} = X'\underline{y} \;\Rightarrow\; \hat{\boldsymbol{\beta}}^{\text{Ridge}} = (X'X + \lambda I_p)^{-1}X'\underline{y} \tag{$***$}$$

- The solution adds a positive constant to the diagonal of $X'X$ before inversion. This makes the problem nonsingular, even if $X'X$ is not full rank. This was the main motivation when it was first introduced in statistics.

- We know that OLS estimates are scale invariant, indeed, consider the rescaled design matrix $Z = XR$, then:

$$\hat{\boldsymbol{\beta}}_X^{\text{OLS}} = (X'X)^{-1}X'\underline{y}, \quad \text{and:} \tag{3}$$

$$\begin{aligned} \hat{\boldsymbol{\beta}}_Z^{\text{OLS}} &= \left((XR)'XR\right)^{-1}(XR)'\underline{y} = \left(R'X'XR\right)^{-1}R'X'\underline{y} &(4)\\ &= R^{-1}(X'X)^{-1}(R')^{-1}R'X'\underline{y} = R^{-1}(X'X)^{-1}X'\underline{y} = R^{-1}\hat{\boldsymbol{\beta}}_X^{\text{OLS}}, &(5) \end{aligned}$$

so that no matter how we rescale the regressors, we obtain the same result.

- However, the Ridge solutions are *not* equivariant under scaling of the inputs, indeed:

$$\hat{\boldsymbol{\beta}}_X^{\text{Ridge}} = (X'X + \lambda I_p)^{-1}X'\underline{y}, \quad \text{and:} \tag{6}$$

$$\begin{aligned} \hat{\boldsymbol{\beta}}_Z^{\text{Ridge}} &= \left(R'X'XR + \lambda I_p\right)^{-1}R'X'\underline{y} &(7)\\ &= \left(R'X'XR + \lambda R'(R')^{-1}R^{-1}R\right)^{-1}R'X'\underline{y} &(8) \end{aligned}$$

$$= \left[R' \left(X'X + \lambda(R')^{-1}R^{-1}\right) R\right]^{-1} R'X'\underline{y} \tag{9}$$

$$= R^{-1} \left(X'X + \lambda(R')^{-1}R^{-1}\right)^{-1} (R')^{-1}R'X'\underline{y} \tag{10}$$

$$\Rightarrow \hat{\boldsymbol{\beta}}_Z^{\text{Ridge}} = R^{-1} \left(X'X + \lambda(R')^{-1}R^{-1}\right)^{-1} X'\underline{y}, \quad \text{hence:} \tag{11}$$

$$\hat{\boldsymbol{\beta}}_Z^{\text{Ridge}} = R^{-1}\hat{\boldsymbol{\beta}}_X^{\text{Ridge}} \iff (R')^{-1}R^{-1} = I_p \iff R'R = I_p, \tag{12}$$

so that the Ridge estimator is scale invariant only when the scale matrix is orthogonal. Hence, one normally standardizes the inputs before fitting a Ridge regression.

- Expression $(***)$ for $\hat{\boldsymbol{\beta}}^{\text{Ridge}}$ makes clear why Ridge regression is biased but has a lower variance than OLS. Consider the case where the noise is such that $\mathbb{E}(\varepsilon \mid X) = 0$ and $\mathbb{V}(\varepsilon \mid X) = \sigma^2 I_N$, then:

$$\mathbb{E}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \mathbb{E}\big[(X'X + \lambda I_p)^{-1}X'\underline{y} \mid X\big] \tag{13}$$

$$= \mathbb{E}\big[(X'X + \lambda I_p)^{-1}X'(X\beta + \varepsilon) \mid X\big] \tag{14}$$

$$= (X'X + \lambda I_p)^{-1}X'X\beta + (X'X + \lambda I_p)^{-1}X'\mathbb{E}(\varepsilon \mid X) \tag{15}$$

$$\Rightarrow \mathbb{E}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = (X'X + \lambda I_p)^{-1}X'X\beta, \quad \text{so that:} \tag{16}$$

$$\mathbb{E}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \beta \iff \lambda = 0; \quad \text{the Ridge estimator is unbiased iff } \lambda = 0. \tag{17}$$

- From the $(X'X + \lambda I_p)^{-1}$ term it is clear that as λ increases, the bias increases. This can be seen from the simpler setup where $N = p$ and $X = I_p$, so that:

$$\hat{\boldsymbol{\beta}}^{\text{OLS}} = \left(I_p' I_p\right)^{-1} I_p' \underline{y} = I_p^{-1} I_p \underline{y} = \underline{y}, \quad \text{and:} \tag{18}$$

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \left(I_p' I_p + \lambda I_p\right)^{-1} I_p' \underline{y} = (I_p + \lambda I_p)^{-1} I_p \underline{y} \tag{19}$$

$$= \begin{pmatrix} \frac{1}{1+\lambda} & 0 & \cdots & 0 \\ 0 & \frac{1}{1+\lambda} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{1+\lambda} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} \tag{20}$$

$$\Rightarrow \hat{\boldsymbol{\beta}}^{\text{Ridge}} = \frac{1}{1+\lambda} \underline{y}, \quad \text{so that } \lambda = 0 \Rightarrow \hat{\boldsymbol{\beta}}^{\text{Ridge}} = \underline{y} = \hat{\boldsymbol{\beta}}^{\text{OLS}}. \tag{21}$$

- Again from the $(X'X+\lambda I_p)^{-1}$, analogous to the $(X'X)^{-1}$ in OLS, it is also clear that the variance is reduced. Putting in light the bias-variance tradeoff. Indeed, first consider the relation between Ridge and OLS:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = (X'X + \lambda I_p)^{-1} X' \underline{y} = (X'X + \lambda I_p)^{-1} X'X (X'X)^{-1} X' \underline{y} \tag{22}$$
$$= (X'X + \lambda I_p)^{-1} X'X \hat{\boldsymbol{\beta}}^{\text{OLS}}, \quad \text{it then becomes that:} \tag{23}$$

$$\mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \mathbb{V}\Big[(X'X + \lambda I_p)^{-1} X'X \hat{\boldsymbol{\beta}}^{\text{OLS}} \mid X\Big] \tag{24}$$
$$= (X'X + \lambda I_p)^{-1} X'X \ \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{OLS}} \mid X\Big) \ X'X (X'X + \lambda I_p)^{-1} \tag{25}$$
$$= \sigma^2 (X'X + \lambda I_p)^{-1} X'X (X'X)^{-1} X'X (X'X + \lambda I_p)^{-1} \tag{26}$$

$$\Rightarrow \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \sigma^2 (X'X + \lambda I_p)^{-1} X'X (X'X + \lambda I_p)^{-1} \tag{27}$$

- Using $W = X'X + \lambda I_p \Rightarrow X'X = W - \lambda I_p$ and therefore:

$$\mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \sigma^2 W^{-1} (W - \lambda I_p) W^{-1} \tag{28}$$
$$= \sigma^2 \left(I_p - \lambda W^{-1}\right) W^{-1} \tag{29}$$

$$\Rightarrow \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \sigma^2 \left(W^{-1} - \lambda W^{-2}\right), \quad \text{taking the difference:} \tag{30}$$

$$\Delta = \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{OLS}} \mid X\Big) - \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) = \sigma^2 \left[(X'X)^{-1} - W^{-1} + \lambda W^{-2}\right] \tag{31}$$

- Using the identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$, we have that:

$$(X'X)^{-1} - W^{-1} = (X'X)^{-1} \left[W - X'X\right] W^{-1} = \lambda (X'X)^{-1} W^{-1}, \quad \text{yielding:} \tag{32}$$

$$\Delta = \sigma^2 \left[\lambda (X'X)^{-1} W^{-1} + \lambda W^{-2}\right] = \sigma^2 \lambda \left[(X'X)^{-1} W^{-1} + W^{-2}\right] \tag{33}$$

- Because W is positive definite, W^{-2} is also p.d. (the eigenvalues of W are $d_i > 0 \ \forall i$, then the eigenvalues of W^{-2} are $\frac{1}{d_i^2} > 0 \ \forall i$).

- As $X'X$ is p.d., $W = X'X + \lambda I_p$ is p.d.
- Now: $(X'X)W = (X'X)^2 + \lambda X'X$, and:

$$W(X'X) = (X'X)^2 + \lambda(X'X) = (X'X)W. \tag{34}$$

- Because $(X'X)$ and W are p.d. and commute, we have that $(X'X)W$ is p.d., as well as its inverse.
- Because the sum of two p.d. matrices is p.d., it follows that Δ is p.d., i.e. $\mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{OLS}} \mid X\Big) - \mathbb{V}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}} \mid X\Big) > 0$, so that for every single coefficient β_h, the Ridge estimator has strictly smaller variance than the OLS estimator.
- In $(*)$ and $(**)$, notice that the intercept β_0 has been left out of the penalty term. The intuition is the following, consider the simple model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad x_i \in \mathbb{R}. \tag{35}$$

- Suppose we shift the regressor by a constant term c, to obtain $x^* = x + c$, to keep the same predicted values $\hat{y}$, we need: $\beta_0^* = \beta_0 - \beta_1 c$.
- However, if we penalize the intercept, the penalty goes from $\lambda\beta_1^2 + \lambda\beta_0^2$ to $\lambda\beta_1^2 + \lambda(\beta_0 - \beta_1 c)^2$, so that $\hat{\boldsymbol{\beta}}^{\text{Ridge}}$ will depend on the mean of the data. This is an undesirable property because the model should be invariant to simple location shifts of the inputs.
- On the other hand, if we do not penalize the intercept, the penalty stays $\lambda\beta_1^2$ and the intercept can absorb the shift c and $\hat{\boldsymbol{\beta}}_1^{\text{Ridge}}$ remains the same.

- In practice, it can be shown that the solution to $(*)$ can be separated in two parts:
  (i) Each x_{ij} gets replaced by $x_{ij} - \bar{x}_j$ and β_0 estimated by $\bar{y}$.
  (ii) Remaining coefficients $\{\beta_j\}_{j=1}^{p}$ get estimated by a Ridge regression without intercept.
- Indeed, consider the following problem:

$$\hat{\boldsymbol{\beta}}^c = \arg\min_{\beta^c} \left\{ \sum_{i=1}^{N} \left(y_i - \beta_0^c - \sum_{j=1}^{p} (x_{ij} - \bar{x})\beta_j^c \right)^2 + \lambda \sum_{j=1}^{p} {\beta_j^c}^2 \right\} \qquad (**_c)$$

- The original linear predictor is:

$$\beta_0 + \sum_{j=1}^{p} x_{ij}\beta_j = \beta_0 + \sum_{j=1}^{p} (z_{ij} + \bar{x}_j)\beta_j, \quad \text{where } z_{ij} = x_{ij} - \bar{x}_j. \qquad (36)$$

- Define β_0^c and β_j^c as: $\beta_0^c = \beta_0 + \sum_{j=1}^{p} \bar{x}_j\beta_j;\ \ \beta_j^c = \beta_j\ \forall j = 1, \ldots, p$. The original problem is:

$$\min_{\beta} \left\{ \sum_{i=1}^{N} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\} \qquad (37)$$

$$\iff \min_{\beta} \left\{ \sum_{i=1}^{N} \left(y_i - \left[\beta_0 + \sum_{j=1}^{p} \bar{x}_j\beta_j \right] - \sum_{j=1}^{p} z_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\} \qquad (38)$$

$$\iff (**_c) \qquad (39)$$

- Then, the correspondence is the following:

$$\hat{\beta}_j^c = \hat{\beta}_j^{\text{Ridge}} \quad \forall j = 1, \ldots, p \qquad (40)$$

$$\hat{\beta}_0^c = \hat{\beta}_0^{\text{Ridge}} + \sum_{j=1}^{p} \bar{x}_j \hat{\beta}_j^{\text{Ridge}} \qquad (41)$$

- The solution to this modified criterion is easily obtained:

$$
\begin{aligned}
\partial_{\beta_0^c}\left(\sum_{i=1}^{N}\left(y_i-\beta_0^c-\sum_{j=1}^{p}(x_{ij}-\bar{x})\beta_j^c\right)^2\right)\Bigg|_{\beta_0^c=\hat{\beta}_0^c} &= 0 && (42)\\
\iff -2\sum_{i=1}^{N}\left(y_i-\hat{\beta}_0^c-\sum_{j=1}^{p}(x_{ij}-\bar{x})\beta_j^c\right) &= 0 && (43)\\
\iff \sum_{i=1}^{N}y_i-N\hat{\beta}_0^c-\sum_{i=1}^{N}\sum_{j=1}^{p}(x_{ij}-\bar{x})\beta_j^c &= 0 && (44)\\
\iff \bar{y}-\hat{\beta}_0^c &= 0 \quad \text{as} \sum_{i=1}^{N}(x_{ij}-\bar{x}) = N\bar{x}-N\bar{x}=0 && (45)\\
\iff \hat{\beta}_0^c &= \bar{y}, && (46)
\end{aligned}
$$

so that the intercept is the mean of the response.

- Substituting into the objective function:

$$
\sum_{i=1}^{N}\left(y_i-\bar{y}-\sum_{j=1}^{p}z_{ij}\beta_j^c\right)^2+\lambda\sum_{j=1}^{p}\beta_j^{c\,2} \tag{47}
$$

- Let Z be the centered design matrix, then:

$$
\begin{aligned}
\hat{\boldsymbol{\beta}}^c &= \arg\min_{\beta^c}\left\{\left\|\underline{y}^c-Z\beta^c\right\|_2^2+\lambda\left\|\beta^c\right\|_2^2\right\} && (48)\\
&= (Z'Z+\lambda I_p)^{-1}Z'\underline{y}^c, \quad \text{where } \underline{y}^c=\underline{y}-\bar{y}. && (49)
\end{aligned}
$$

- Another standard way to see Ridge regression is as a Bayes estimator. Consider the prior assumptions: $\beta \sim \mathcal{N}\big(0, \frac{1}{\lambda}I_p\big)$, meaning that we suspect the true effects to be small and the predictors are independent in their effects (with equal magnitude).

- Then, consider the model: $\underline{y} \mid \beta \sim \mathcal{N}(X\beta, I)$, so that the observed outcomes are a linear combination plus some random noise.

- Then, the density of β is:

$$p(\beta) = \frac{1}{\sqrt{(2\pi)^p \det\left(\frac{1}{\lambda} I_p\right)}} \exp\left(-\frac{1}{2} \beta' \left(\frac{1}{\lambda} I_p\right)^{-1} \beta \right) \propto \exp\left(-\frac{1}{2} \beta' \lambda I \beta \right) = \exp\left(-\frac{\lambda}{2} \|\beta\|_2^2 \right) \tag{50}$$

- The likelihood for observed data is:

$$p(\underline{y} \mid \beta) = \frac{1}{\sqrt{(2\pi)^N \det(I_N)}} \exp\left(-\frac{1}{2} (\underline{y} - X\beta)'(\underline{y} - X\beta) \right) \propto \exp\left(-\frac{1}{2} \|\underline{y} - X\beta\|_2^2 \right) \tag{51}$$

- The posterior is then:

$$\begin{aligned} p(\beta \mid \underline{y}) &= p(\underline{y} \mid \beta)\, p(\beta)/p(\underline{y}) && (52) \\ &\propto p(\underline{y} \mid \beta)\, p(\beta) && (53) \\ &= \exp\left(-\frac{1}{2} \|\underline{y} - X\beta\|_2^2 - \frac{\lambda}{2} \|\beta\|_2^2 \right) && (54) \end{aligned}$$

- The MAP estimator is the value of β maximizing the posterior probability:

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{\text{MAP}} &= \arg\max_\beta \left\{ p(\beta \mid \underline{y}) \right\} \quad \text{(Mode)} && (55) \\ &= \arg\max_\beta \left\{ \exp\left(-\frac{1}{2} \|\underline{y} - X\beta\|_2^2 - \frac{1}{2} \lambda \|\beta\|_2^2 \right) \right\} && (56) \\ &= \arg\min_\beta \left\{ \exp\left(\frac{1}{2} \|\underline{y} - X\beta\|_2^2 + \frac{1}{2} \lambda \|\beta\|_2^2 \right) \right\} && (57) \end{aligned}$$

$$\Rightarrow \hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg\min_\beta \left\{ \|\underline{y} - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\} = \hat{\boldsymbol{\beta}}^{\text{Ridge}} \tag{58}$$

- Hence, the Ridge estimate is the mode of the posterior distribution. Because the distribution is Gaussian, it is also the posterior mean.
- Consider the centered design matrix $X \in \mathbb{R}^{N\times p}$, its SVD has the form:

$$X = U\Sigma V', \quad \text{where:} \tag{59}$$

* $U \in \mathbb{R}^{N\times p}$ and $V \in \mathbb{R}^{p\times p}$ are orthogonal matrices.
* $\Sigma \in \mathbb{R}^{p\times p}$ with diagonal values $d_1 \geq d_2 \geq \cdots \geq d_p \geq 0$ the singular values of X (diagonal matrix).

- If $\exists j : d_j = 0$, X is singular.
- Using the SVD, the fitted values from least squares write:

$$\begin{aligned}
\underline{\hat{y}}^{\text{ls}} = X\hat{\boldsymbol{\beta}}^{\text{ls}} &= X(X'X)^{-1}X'\underline{y} && (60)\\
&= U\Sigma V'\left(V\Sigma U'U\Sigma V'\right)^{-1}V\Sigma U'\underline{y} && (61)\\
&= U\Sigma V'\left(V\Sigma\Sigma V'\right)^{-1}V\Sigma U'\underline{y} && (62)\\
&= U\Sigma V'V\Sigma^{-2}V'V\Sigma U'\underline{y} && (63)\\
&= U\Sigma\Sigma^{-2}\Sigma U'\underline{y} && (64)
\end{aligned}$$

$$\Rightarrow X\hat{\boldsymbol{\beta}}^{\text{ls}} = UU'\underline{y}, \tag{65}$$

where $U'\underline{y}$ are the coordinates of $\underline{y}$ with respect to the orthonormal basis U.

- On the other hand, the Ridge solutions are:

$$
\begin{aligned}
\underline{\hat{y}}^{\text{Ridge}} = X\hat{\boldsymbol{\beta}}^{\text{Ridge}} &= X(X'X + \lambda I_p)^{-1}X'\underline{y} &(66)\\
&= U\Sigma V'\left(V\Sigma^2V' + \lambda I_p\right)^{-1}V\Sigma U'\underline{y} &(67)\\
&= U\Sigma V'\left[V\Sigma^2V' + \lambda VV'\right]^{-1}V\Sigma U'\underline{y} &(68)\\
&= U\Sigma V'\left[V\left(\Sigma^2 + \lambda I_p\right)V'\right]^{-1}V\Sigma U'\underline{y} &(69)\\
&= U\Sigma V'V\left(\Sigma^2 + \lambda I_p\right)^{-1}V'V\Sigma U'\underline{y} &(70)\\
&= U\Sigma\left(\Sigma^2 + \lambda I_p\right)^{-1}\Sigma U'\underline{y} &(71)
\end{aligned}
$$

$$
= U\Sigma \begin{pmatrix} d_1^2+\lambda & 0 & \cdots & 0 \\ 0 & d_2^2+\lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_p^2+\lambda \end{pmatrix}^{-1} \Sigma U'\underline{y} \tag{72}
$$

$$
= U \begin{pmatrix} \dfrac{d_1^2}{d_1^2+\lambda} & 0 & \cdots & 0 \\ 0 & \dfrac{d_2^2}{d_2^2+\lambda} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \dfrac{d_p^2}{d_p^2+\lambda} \end{pmatrix} U'\underline{y} \tag{73}
$$

$$
= \sum_{j=1}^{p} u_j \frac{d_j^2}{d_j^2+\lambda} u_j'\underline{y}, \tag{74}
$$

where u_j is the j-th column of U.

- Since $\lambda \geq 0$, $d_j^2/(d_j^2+\lambda) \leq 1$.

- Hence, like linear regression, ridge regression computes the coordinates of $\underline{y}$ w.r.t. the orthonormal basis U. It then shrinks these coordinates by $d_j^2/(d_j^2+\lambda)$.
- This means that a greater amount of shrinkage is applied to the coordinates of basis vectors with smaller d_j^2. To understand what this means, take the sample covariance matrix $S = X'X/N$, we have that:

$$X'X = V\Sigma U'U\Sigma V' = V\Sigma^2 V', \quad \text{which is the eigendecomposition of } X'X. \tag{75}$$

- The associated eigenvectors are the columns, v_j, of V. They are also called the principal components directions of X.
- We know that the 1st PC direction v_1 is such that $z_1 = Xv_1$ has the largest sample variance, equal to:

$$\begin{aligned}
\mathbb{V}(z_1) = \mathbb{V}(Xv_1) &= \mathbb{V}(U\Sigma V' v_1) && (76)\\
&= \mathbb{V}(U\Sigma e_1), \quad \text{where } e_1 = (1,0,\ldots,0)' && (77)\\
&= \mathbb{V}(Ud_1e_1) = \mathbb{V}(d_1u_1) && (78)\\
&= \frac{1}{N}d_1^2 u_1' u_1 = \frac{d_1^2}{N} && (79)
\end{aligned}$$

- Subsequent PC z_j have maximum variance d_j^2/N, subject to being orthogonal to the previous ones. Hence, small singular values d_j correspond to directions in the column space of X having small variance. Ridge regression shrinks these directions the most.

- We now derive the Mean Squared Error of the Ridge estimator:

$$\begin{aligned}
\operatorname{MSE}\left(\hat{\boldsymbol{\beta}}^{\text{Ridge}}\right) &= \mathbb{E}\left[\left(\hat{\boldsymbol{\beta}}^{\text{Ridge}} - \beta\right)'\left(\hat{\boldsymbol{\beta}}^{\text{Ridge}} - \beta\right)\right] && (80)\\
&= \mathbb{E}\left[\left(W_\lambda\hat{\beta} - \beta\right)'\left(W_\lambda\hat{\beta} - \beta\right)\right], \quad \text{where } \hat{\beta} \equiv \hat{\boldsymbol{\beta}}^{\text{ls}} && (81)
\end{aligned}$$

with $W_\lambda = (X'X + \lambda I_p)^{-1}X'X$. Expanding the quadratic form and collecting the cross terms (which telescope), we are left with:

$$\operatorname{MSE}\left(\hat{\boldsymbol{\beta}}^{\text{Ridge}}\right) = \mathbb{E}\left[\left(\hat{\beta} - \beta\right)' W_\lambda' W_\lambda \left(\hat{\beta} - \beta\right)\right] + \beta'\left(W_\lambda - I_p\right)'\left(W_\lambda - I_p\right)\beta \tag{82}$$

Note that for a multivariate random variable $\varepsilon \sim \mathcal{N}(\mu_\varepsilon, \Sigma_\varepsilon)$, we have that the expectation of a quadratic form is:

$$\mathbb{E}(\varepsilon'\Lambda\varepsilon) = \operatorname{Tr}(\Lambda\Sigma_\varepsilon) + \mu_\varepsilon'\Lambda\mu_\varepsilon, \quad \text{then, it follows that:} \tag{83}$$

as $\hat{\beta} - \beta \sim \mathcal{N}\big(0, \sigma^2(X'X)^{-1}\big)$ we get:

$$\text{MSE}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}}\Big) = \text{Tr}\big(W_\lambda' W_\lambda \sigma^2 (X'X)^{-1}\big) + \beta' \left(W_\lambda - I_p\right)' \left(W_\lambda - I_p\right) \beta \tag{84}$$

$$= \sigma^2 \text{Tr}\big(W_\lambda' W_\lambda (X'X)^{-1}\big) + \beta' \left(W_\lambda - I_p\right)' \left(W_\lambda - I_p\right) \beta \tag{85}$$

$$\Rightarrow \text{MSE}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}}\Big) = \sigma^2 \text{Tr}\big(W_\lambda (X'X)^{-1} W_\lambda'\big) + \beta' \left(W_\lambda - I_p\right)' \left(W_\lambda - I_p\right) \beta \tag{86}$$

(using $\text{Tr}(ABC) = \text{Tr}(CAB) = \text{Tr}(BCA)$).

- Because $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X'X)^{-1})$, its bias term is zero, so that:

$$\text{MSE}(\hat{\beta}) = \sigma^2 \text{Tr}\big[(X'X)^{-1}\big] \tag{87}$$

- Define $g(\lambda) = \text{MSE}(\hat{\beta}) - \text{MSE}\Big(\hat{\boldsymbol{\beta}}^{\text{Ridge}}(\lambda)\Big)$, we already know that $g(0) = 0$. If we can show that $g'(0) > 0$, then for $\lambda > 0$ small enough, we have $g(\lambda) > 0$.
- Let $V(\lambda) = \sigma^2 \text{Tr}\big(W_\lambda (X'X)^{-1} W_\lambda'\big)$:

$$V(\lambda) = \sigma^2 \text{Tr}\big((X'X + \lambda I_p)^{-1} X'X (X'X)^{-1} (X'X)(X'X + \lambda I_p)^{-1}\big) \tag{88}$$

$$= \sigma^2 \text{Tr}\big((X'X + \lambda I_p)^{-1} X'X (X'X + \lambda I_p)^{-1}\big) \tag{89}$$

Clearly: $\partial_\lambda \left((X'X + \lambda I_p)^{-1}\right)\big|_{\lambda=0} = -(X'X + \lambda I_p)^{-2}\big|_{\lambda=0} = -(X'X)^{-2}$, hence:

$$V'(0) = \sigma^2 \text{Tr}\big(-(X'X)^{-2}(X'X)(X'X)^{-1} - (X'X)^{-1}(X'X)(X'X)^{-2}\big) \tag{90}$$

$$\Rightarrow V'(0) = -2\sigma^2 \text{Tr}\big((X'X)^{-2}\big) \tag{91}$$

- Define $B(\lambda) = \beta'(W_\lambda - I_p)'(W_\lambda - I_p)\beta$, we have that:

$$
\begin{aligned}
\partial_\lambda W_\lambda|_{\lambda=0} &= \partial_\lambda \left[(X'X + \lambda I_p)^{-1} X'X\right]\Big|_{\lambda=0} && (92)\\
&= -(X'X)^{-2}(X'X) = -(X'X)^{-1}, \quad \text{then:} && (93)
\end{aligned}
$$

$$
\begin{aligned}
\partial_\lambda \left[(W_\lambda - I_p)'(W_\lambda - I_p)\right]\Big|_{\lambda=0} &= \partial_\lambda W'_\lambda\Big|_{\lambda=0} \left(W_\lambda|_{\lambda=0} - I_p\right) && (94)\\
&\quad + \left(W_\lambda|_{\lambda=0} - I_p\right)' \partial_\lambda W_\lambda|_{\lambda=0} && (95)\\
&= -(X'X)^{-1}(I_p - I_p) && (96)\\
&\quad + (I_p - I_p)' \left(-(X'X)^{-1}\right) && (97)\\
&= 0, \quad \text{hence:} && (98)
\end{aligned}
$$

$$
g'(0) = -V'(0) = 2\sigma^2 \mathrm{Tr}\big((X'X)^{-2}\big) > 0 \quad \text{as } X'X \text{ is p.d.} \tag{99}
$$

- Hence, we can conclude that $\exists\, \lambda > 0 : \mathrm{MSE}\Big(\hat{\boldsymbol{\beta}}^{\mathrm{Ridge}}\Big) < \mathrm{MSE}(\hat{\beta})$. Again, illustrating the bias-variance tradeoff.
- The problem is that this tradeoff depends on unknown quantities β and σ^2.
- A popular strategy is to choose the penalty parameter yielding a "good" but parsimonious model through the use of an information criterion.
- Recall the general expression for the AIC criterion:

$$\mathrm{AIC}(\lambda) = 2\,\mathrm{df}(\lambda) - 2\log \hat{L} \tag{100}$$

$$\mathrm{AIC}(\lambda) = 2\sum_{j=1}^{p}\frac{d_j^2}{d_j^2+\lambda} + 2N\log\Big(\sqrt{2\pi}\,\hat{\sigma}(\lambda)\Big) + \frac{\sum_{i=1}^{N}\Big(y_i - x_i'\hat{\beta}(\lambda)\Big)^2}{\hat{\sigma}^2(\lambda)} \tag{101}$$

Recall that $\mathcal{L}(\beta) = \prod_{i=1}^{N}\left\{\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\frac{(y_i - x_i'\beta)^2}{\sigma^2}\right)\right\}$ (for a normal model)

$$= \left(\sigma\sqrt{2\pi}\right)^{-N}\exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{N}(y_i - x_i'\beta)^2}{\sigma^2}\right) \tag{102}$$

$$\Rightarrow \ell(\beta) = -N\log\Big(\sqrt{2\pi}\,\sigma\Big) - \frac{1}{2}\frac{\sum_{i=1}^{N}(y_i - x_i'\beta)^2}{\sigma^2} \tag{103}$$

- The value of λ minimizing $\mathrm{AIC}(\lambda)$ corresponds to the "optimal" balance of model complexity and overfitting.
- Instead of choosing λ to balance model fit with model complexity, CV requires it to yield a model with good prediction performance.
- For LOOCV, the procedure is the following:
  - (i) Define a range of interest for λ.
  - (ii) Divide the data into training: $\{x_\ell, y_\ell\}_{\ell=1}^{N},\ \ell \neq i$ and test (x_i, y_i).
  - (iii) $\forall\lambda$ in the grid, compute:

$$\hat{\boldsymbol{\beta}}_{-i}^{\mathrm{Ridge}}(\lambda) = \left(X_{-i}'X_{-i} + \lambda I_p\right)^{-1} X_{-i}'\underline{y}_{-i} \tag{104}$$

  - (iv) Evaluate the performance on the corresponding test data $L\Big(y_i, x_i'\hat{\boldsymbol{\beta}}_{-i}^{\mathrm{Ridge}}(\lambda)\Big)$.
  - (v) Repeat (ii)-(iv) $\forall i = 1, \ldots, N$.

(vi) Average these performances:

$$\frac{1}{N}\sum_{i=1}^{N} L\Big(y_i, x_i'\hat{\boldsymbol{\beta}}_{-i}^{\text{Ridge}}(\lambda)\Big) = \text{LOOCV}(\lambda) \tag{105}$$

(vii) The value of λ minimizing LOOCV(λ) is the value of choice.

- This procedure is straightforwardly adapted to K-fold CV.
- Using the squared loss:

$$\frac{1}{N}\sum_{i=1}^{N} L\Big(y_i, x_i'\hat{\boldsymbol{\beta}}_{-i}^{\text{Ridge}}(\lambda)\Big) = \frac{1}{N}\sum_{i=1}^{N}\Big(y_i - x_i'\hat{\boldsymbol{\beta}}_{-i}^{\text{Ridge}}(\lambda)\Big)^2, \tag{106}$$

it turns out that the prediction performance can be expressed analytically in terms of known quantities derived from X and $\underline{y}$.

- That is, the prediction performance for given λ can be assessed directly from the data without the recalculation of N LOO ridge estimators, which is a considerable gain in terms of computation.
- We actually derived that formula in the CV section for any additive model of the form $y_i = f(x_i) + \varepsilon_i$. In our case:

$$\text{LOOCV}(\lambda) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{y_i - x_i'\hat{\boldsymbol{\beta}}^{\text{Ridge}}(\lambda)}{1-[H_\lambda]_{ii}}\right)^2, \tag{107}$$

where $[H_\lambda]_{ii}$ is the i-th diagonal element of the Ridge hat matrix:

$$H_\lambda = X(X'X + \lambda I_p)^{-1}X' \tag{108}$$